

Work Experience

Software Engineer, Stairwell. *Sunnyvale, CA.*

Feb 2024 - Present

Stairwell is a cybersecurity startup that measures sightings of files on an organization's assets, continuously scans them against AV engines and YARA rules, extracts indicators of compromise (hashes, hostnames, IPs) to detect live and retroactive threats, and provides tools to triage and run them to ground.

- Developed gRPC/Go microservices on the Infrastructure team, primarily focused on the ingestion, indexing, scanning, and querying of a 9 billion file corpus, under strict SLOs in a cybersecurity context.
- Refactored and improved test suites to reduce flakes and decrease median time-to-merge by 5x.
- Built stable customer-facing APIs and integrations with other security tools (Splunk, CrowdStrike) plugged into performant indexes on our file, networking, and UGC data.
- Implemented a unified data model to define relationships between files, network entities, scan results, etc. and allow nuanced graph queries with propagating tags.
- Centralized search in a query mixer, with database backends using Spanner, Bigtable, and Elasticsearch.
- Augmented our Elasticsearch indexer with rate limiting and scheduled work queues to ensure peaks of 40k attempted indexing ops/sec do not compromise search latency.
- Executed zero-downtime migrations of critical systems, including our auth service and entire 600TiB corpus.
- Triageed and resolved critical issues, built tools to reduce toil, improved monitoring, and wrote robust postmortems on an oncall rotation.
- Led the project to instrument our entire system with distributed tracing using OpenTelemetry, along with a self-hosted Grafana Tempo backend and dev tooling leading to adoption on the frontend and backend teams.

Platform Engineer, Emerald Cloud Lab. *Remote.*

June 2023 - Jan 2024

- Worked as an engineer on the Platform team, maintaining the large database and various backend services that orchestrate the cloud laboratory and support experimental development processes of biologists and chemists.
- Built an optimized REST API using Go to greatly reduce latency for common experimental procedures like HPLC.
- Designed and implemented an orchestration system for pools of Mathematica kernels hosted in AWS Fargate, allowing on-demand experimental operations using Wolfram Language to bypass minutes of startup time.
- Incorporated a custom linter for scientific Wolfram Language code as a CI check on pull requests, using GitHub webhooks and Python serverless runners on AWS Lambda.
- Fixed critical platform bugs during the 24/7 operation of the lab on an oncall rotation.

Software Developer, AI Learners. *Remote.*

Jan 2021 - Dec 2022

- Worked as a full-stack developer building a modular framework for math and logic web games using React and TypeScript, eventually supporting over 55 different games with a focus on inclusive design.
- Adopted the additional role of a project manager for one summer, planning projects and timelines, and coordinating tasks between other developers in a remote work setting.
- Ensured compatibility with assistive technologies like screen readers, adhering to WCAG standards.
- Practiced end-to-end testing with Cypress to implement robust front-end components compatible across browsers and responsive for both mobile and desktop devices.

Teaching Assistant, Cornell University Computing and Information Science. *Ithaca, NY.*

Feb 2021 - May 2023

- Facilitated lab sessions of up to 30 students, held office hours, and communicated technical concepts to students learning Java, JavaScript, HTML/CSS, game development and user-centered design.

Education

Cornell University, College of Arts and Sciences, *Ithaca, NY.*

- Master of Professional Studies in Information Science, Graduated May 2023. GPA: 4.0.
- Bachelor of Arts in Information Science, Graduated December 2022. GPA: 3.9, Dean's List.

Relevant Coursework: Advanced Game Development, Functional Programming, Object-Oriented Programming, Data Structures, Data Science, Machine Learning, Natural Language Processing, Computer Vision, Web Design & Development, Human-Computer Interaction Design, Introduction to Rapid Prototyping.



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Mountain View, CA.

Technical Skills

- **Programming:** Proficient in Go, Python, TypeScript, Java, SQL. Experience with Rust, C#, OCaml, C++.
- **DevOps:** Proficient in Git, Prometheus, Grafana. Experience with Bazel, Docker, Temporal, Spinnaker.
- **Cloud:** Proficient in Kubernetes, Terraform, GCP (Spanner, Bigtable, PubSub, GKE, GCE). Experience with AWS.
- **Web Development:** Proficient in HTML, CSS, React, Vue. Experience with PHP.
- **Data Science:** Proficient in NumPy, pandas, R, scikit-learn. Experience with PyTorch.